CALORIMETRY

OBJECTIVES

- Determine the specific heat of a metal to determine its identity
- Use the specific heat of water to determine the caloric content of food

INTRODUCTION

The First Law of Thermodynamics states that the heat energy lost by one body is gained by another body. Chemical reactions almost always involve a transfer of energy between the reaction system and its surroundings. Thermodynamics is the study of this energy transfer. Energy transfer as heat occurs whenever there is a temperature difference between a system and its surroundings. Energy in the form of heat flows spontaneously from an object at a higher temperature to one at a lower temperature until thermal equilibrium is reached. The energy transferred as heat is represented by the symbol ***q***.

Chemical reactions can both release energy as heat into the surroundings and also take in heat from the surroundings. A reaction that releases energy is called an **exothermic** reaction and *q* has a negative sign. Alternatively, when energy as heat is transferred from the surroundings to the reaction, it is **endothermic** and *q* is positive.

Thermite reactions (Figure 1) are one of the most exothermic reactions known. Thermite was used in World War II to disable enemy artillery pieces without the need for an explosion. Thermite is so exothermic that it would weld the metal of the artillery barrel shut making loading the weapon impossible. Being able to do this quickly and quietly was a real advantage.



Figure 1: The thermite reaction between powdered aluminum metal and iron (III) oxide.

Heat is usually measured in calories or joules. The calorie is defined as the amount of energy to increase the temperature of 1 gram of water one degree Celsius. The joule (J), on the other hand, is the amount of kinetic energy required to carry a 2 kg object moving 1 meter per second. One

calorie is equivalent to 4.184 joules. A kilojoule, or kJ, is 1000 joules and a kilocalorie, or kcal, is 1000 calories (or 1 food Calorie).

Energy changes in chemical reactions are often in the range of 100-500kJ, or 25-125 kcal.

USING TEMPERATURE CHANGES TO MEASURE HEAT

The kinetic energy flowing into or out of a material as heat can be measured by monitoring the resulting temperature change, ΔT, and knowing how many joules or calories are required to warm the material by one degree. For example, at ordinary laboratory temperatures, water requires 4.184 J to warm 1 gram of water by 1ºC. We say that the specific heat (C_s) of water is $4.18\ \frac{J}{g\times ℃}$.

For example, if an aqueous reaction warms 150 mL of its solution by 12ºC, the reaction can be said to have released $\left(4.18\ \frac{J}{g\times ℃}\right)(150\ g)(12℃) = 7500\ J = 7.5\ kJ$ of heat into the solution. Here we have assumed the solution is dilute enough that its specific heat is the same as for pure water.

To calculate either heat absorbed or heat released from the surroundings the formula is:

$$q_{solution} = C_s \times m \times \Delta T$$

$$C_s = specific\ heat\ of\ the\ material$$

$$m = mass\ of\ the\ material$$

$$\Delta T = T_{final} - T_{initial}$$

If looking at a chemical reaction and the surrounding solution it is important to realize that:

$$q_{reaction} = -q_{solution}$$

Or for Part A of today's lab

$$q_{metal} = -q_{water}$$

meaning the change in heat of the solution must be equal and opposite in sign from the change in the heat of the system.

USING SPECIFIC HEAT TO DETERMINE THE IDENTITY OF A METAL

Half of this lab will consist of you measuring the specific heat of a metal and then comparing that to know specific heats to determine its identity. To make accurate measurements of heat transfer and to prevent heat loss to the surroundings, an insulating device known as a ***calorimeter***

is used. In this lab a copper calorimeter will be used. The specific heats for some common substances are found in table 1 below. Compare the heat capacities of concrete and wood for example. It takes about twice as much heat to raise the temperature of wood then concrete. This is why walking on concrete on a hot day feels hotter than walking on wood.

Substance	**Specific Heat** $\left(\frac{J}{g\times℃}\right)$
Water	$4.184 = 1.00\ \frac{cal}{g\times℃}$
Copper	0.385
Aluminum	0.902
Iron	0.451
Lead	0.129
Nickel	0.444
Zinc	0.385
Tin	0.222
Wood	1.76
Concrete	0.88

Table 1: Specific heats of common materials

FOOD AS FUEL

The second part of this lab will consist of measuring the heat energy released from the burning of food. The food we eat constitutes the fuel our bodies need to maintain body temperature, accomplish numerous physiological functions, and provides energy. In this lab you will use the specific heat of water and its change in temperature to determine the caloric content in a Fritos™ and Cheetos™.



Figure 2. Two bags of chips. On the left is Fritos corn chips and on right is Cheetos crunchy chips.

PROCEDURE

IMPORTANT: THIS LAB WILL BE DONE IN TWO PARTS, HALF WILL DO PART A FIRST AND THE OTHER HALF WILL DO PART B FIRST.

PART A: MEASURING THE SPECIFIC HEAT OF TWO UNKNOWN METALS TO DETERMINE THEIR IDENTITY. This can be done at your bench.

1. Copy down the following data tables into your notebook.

Unknown Metal #	**Mass of Metal (g)**	$T_{initial\ metal}$	$T_{final\ metal}$	$\Delta T = T_{final} - T_{initial}$

Table 2: Metal data collection

Unknown Metal #	**Volume of Water (mL)**	**Mass of Water** $d = 1g/mL$	$T_{initial\ H2O}$	$T_{final\ H2O}$	$\Delta T = T_{final} - T_{initial}$

Table 3: Water data collection

2. Weigh out roughly 50 g of two unknown metals in two separate weigh boats. Record the accurate mass in Table 2 in your notebook and on your worksheet. **IMPORTANT:** Make sure to **record the unknown number** of the two metals you selected.

3. Fill a 400 mL beaker with about 300 mL of tap water. Pour one of the unknown metals into a large test tube and clamp it in the water so that the metal is completely submerged in the water. Heat the water to a boil using a hot plate. We are going to assume that the boiling water is $\mathbf{94.5}$ ℃ at Durango's elevation. This is $\mathbf{T_{intial}}$ for the metal.

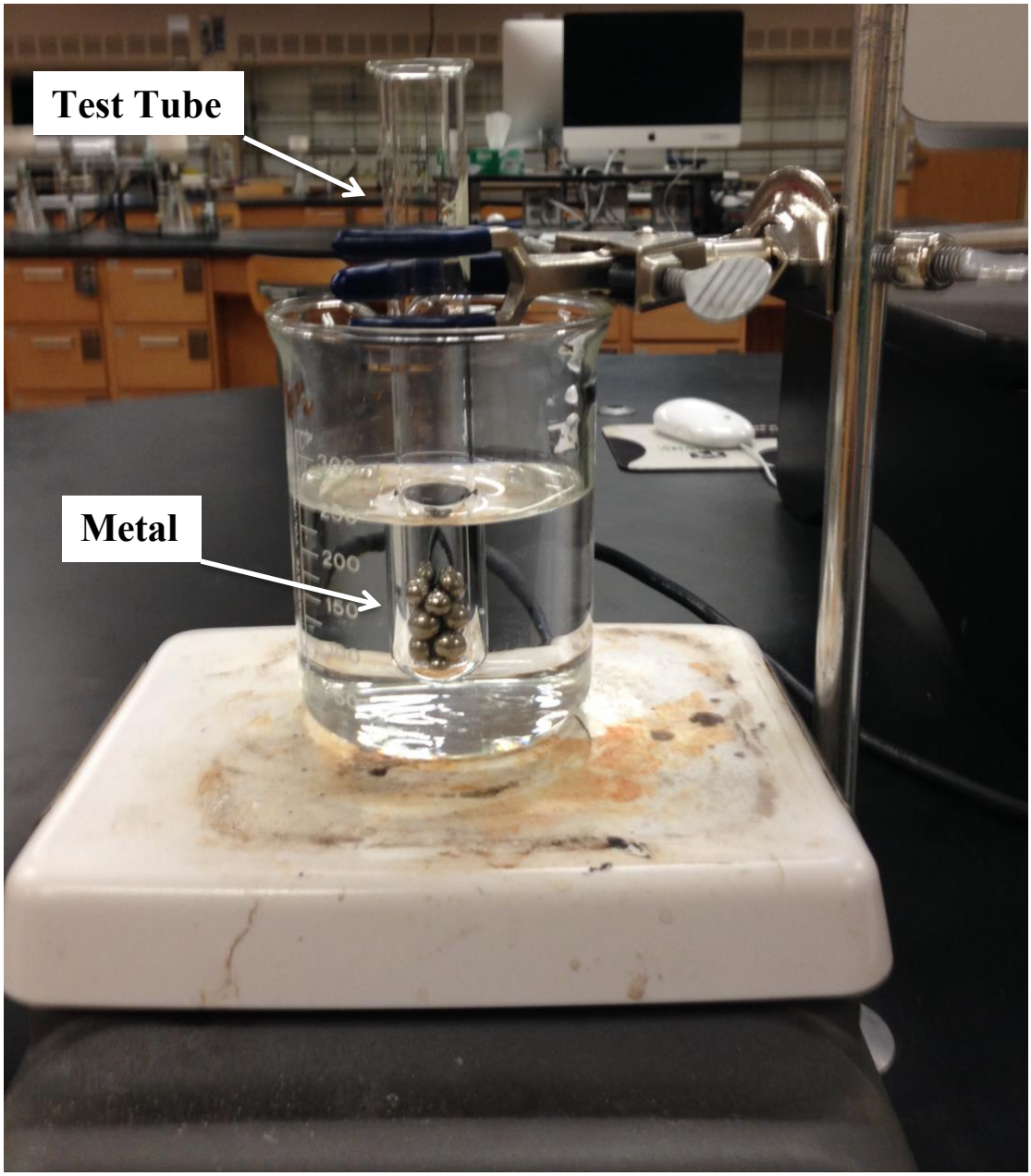


Figure 3. Experimental setup showing a test tube with the unknown metal inside. Using a clamp, the test tube is suspended in a water bath in a beaker. The water should be filled to the half volume of the beaker.

4. While the water is heating to a boil, obtain a calorimeter from the cart. Make sure the calorimeter is clean and completely dry.

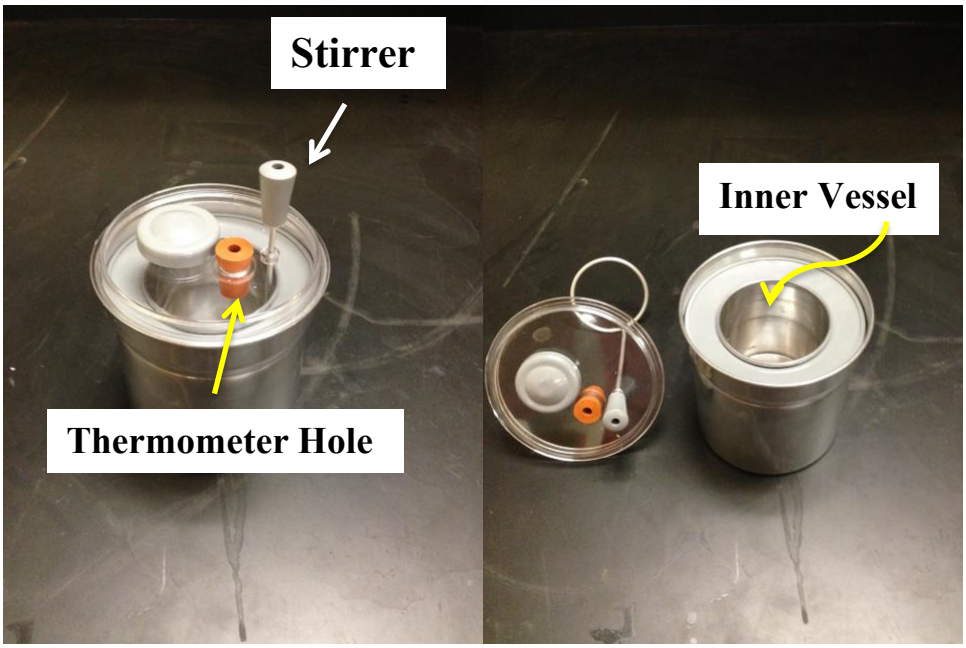


Figure 4. Experimental setup showing the calorimeter.

5. Use a 100 mL graduated cylinder to fill the inner vessel with exactly 100 mL of water and record this volume in the table. Use the density of water ($1\frac{g}{mL}$) to get the mass of water used.

6. Insert a thermometer into the thermometer hole on the calorimeter**. IMPORTANT: Be careful as it might be a tight fit, push the thermometer down only far enough that it will be submerged into the water but not too far as to hit the bottom of the inner vessel.**

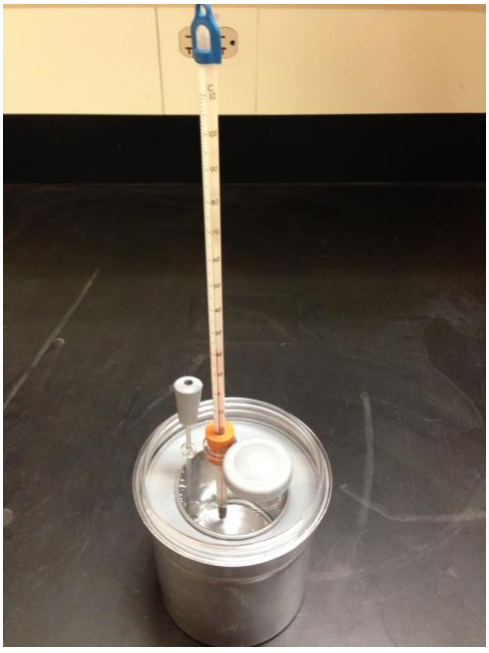


Figure 5. A picture of the assembled calorimeter.

7. Place the top with the thermometer on the calorimeter and give the water inside a stir. Be sure not to stir to vigorously that the water spills out. Wait until the temperature is steady and record this temperature as the $\mathbf{T_{int.}}$ of the water.

8. After the water has been boiling for roughly 5 minutes to ensure that the metal is also at that same temperature; using a test tube clamp, carefully pour the metal into the water in the inner vessel of your calorimeter. Quickly put the top of the calorimeter back on and stir gently.

9. Watch the temperature rise. Once the temperature reaches its maximum record this temperature as $\mathbf{T_{final.}}$ Since we know that both the metal and water are in the calorimeter we can conclude that their final temperatures are the same.

10. You can now calculate the specific heat of the metal.

We know that

$$q_{metal} = -q_{water}$$

so

$$(m_{metal})\left(C_{s(metal)}\right)(\Delta T_{metal}) = -(m_{water})\left(C_{s(water)}\right)(\Delta T_{water})$$

or

$$(m_{metal})\left(C_{s(metal)}\right)(\Delta T_{metal}) = -q_{water}$$

Using the specific heat of water, **4.184 J/g ºC** solve for the specific heat of the metal.

Compare this value to the literature value in Table 1 to identify your unknown.

PUT THE UNKNOWN METALS INTO THE UNKNOWN DRYING BEAKER WITH THE SAME NUMBER AS YOUR UNKNOWN!

PART B: FOOD AS FUEL

IMPORTANT: This should be done in the hood, as gases will be produced.

1. Similar to Part A we will be using the specific heat of water and its temperature change to determine the caloric content in two samples of food. In this part you will also use a calorimeter to get an accurate measurement. However, for ease, an Aluminum can will be used as the calorimeter.

   We can also use the same heat equation from before:

   $$q = C_s \times m \times \Delta T$$

   The normal unit for measuring the energy content in food is called a calorie (cal). The common unit that you see on food labels is actually a kilocalorie or 1000 calories or 1 food Calorie. We will convert the heat energy that is measured from the food sample into calories burned by dividing the heat gained by the water by the change in the mass of the food sample.

2. Copy Table 4 into your notebook.

Food Sample	Initial mass of food holder plus food	Final mass of food holder and food	Change in mass of burned food
Fritos chip			
Cheetos			

Table 4: Food data collection

Food sample	Volume of Water (mL)	Mass of Water $d = 1g/mL$	$T_{initial\ H2O}$	$T_{final\ H2O}$	$\Delta T = T_{final} - T_{initial}$
Fritos chip					
Cheetos					

Table 5: Water data collection

3. Obtain an empty soda can (calorimeter), stir rod, thermometer, and food holder (rubber stopper with a nail) from the cart. Also get a Cheetos™ and Fritos™ food sample.

4. Place one of the food samples on the nail of the food holder and obtain an accurate mass. Record this mass in Table 4.

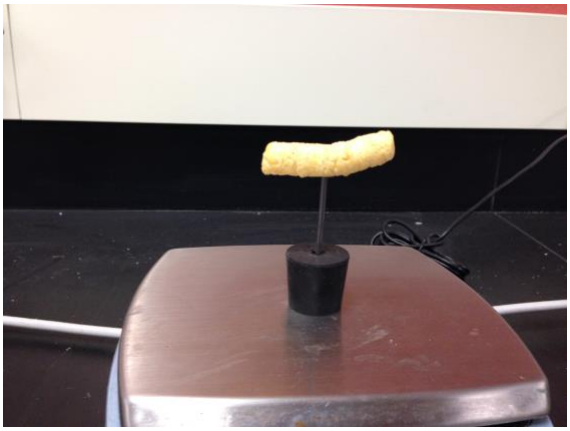


Figure 6. Food sample suspended on a food holder on a balance. The food holder consists of a rubber cork with a nail. The sharp end of the nail allows you to suspend the food for burning.

5. With a graduated cylinder measure exactly 50 mL of water and pour this into the soda can. Bend the tab on the soda can up and insert a glass stir rod through the tab.



Figure 7. Assembled soda can with a metal stir rod going through the tab.

6. Take your soda can and food sample to the hood; and place the can inside the ring stand and place food holder directly under the can as shown below. Add a thermometer into the can and record the temperature of the water as $\mathbf{T_{int.}}$

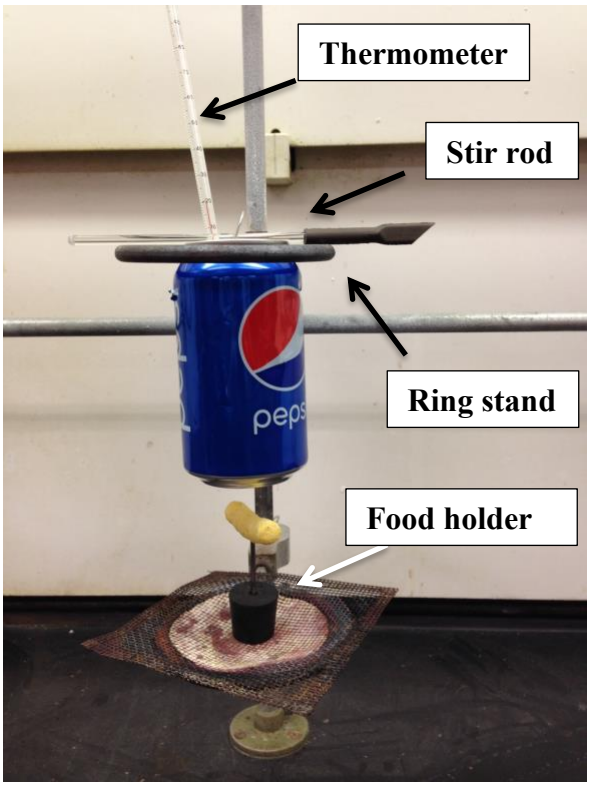


Figure 8. Experimental setup showing the components and placement..

7. Light the food sample on fire with the Bunsen burner. Allow all of the food to burn and then record the final maximum temperature that the water reaches and record this as $\mathbf{T_{final}}$.

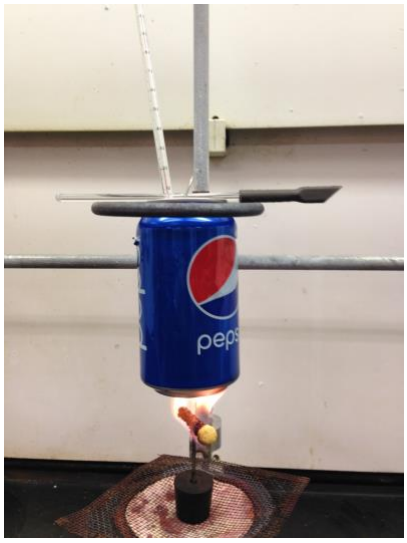


Figure 9. Experimental setup showing the food sample underneath the can as a heat source.

8. Wait for the food holder to cool to room temperature and then weigh the food holder and burned food and record this mass in the table.

9. Clean the bottom of the can and clean any food residue from the holder. Pour out the water and repeat steps 4-9 with the other food sample.

10. Use the heat equation to get the heat released by the food sample:

Heat absorbed by the water = heat released by the food

$$q_{absorbed\ by\ water} = q_{released\ by\ the\ food}$$

Let's use calories to make this calculation easier:

$$q_{solution} = C_s \times m \times \Delta T$$

$$C_s = specific\ heat\ of\ water$$

$$m = mass\ of\ water$$

$$\Delta T = T_{final} - T_{initial}$$

$$4.184\ Joules = 1\ calorie\ (cal)$$

For example:

$$q = \left(1.0\ \frac{cal}{g \times ℃}\right) \times 50.0\ g \times 5.3\ ℃ = 265\ cal$$

Convert the heat gained from calories to food Calories (Cal = kilocalories)

$$265\ cal\ \times \frac{1\ Cal}{1000\ cal} = 0.265\ Cal$$

Calculate the energy content per gram of the food sample

$$\frac{0.265\ Cal}{change\ in\ mass\ of\ food\ sample} = \frac{Cal}{g}$$

11. Calculate the percent (%) error from the actual value on the food bag.

$$\%\ error = \frac{experimental\ value - actual\ value}{actual\ value} \times 100\%$$

IMPORTANT: CLEAN THE CAN, CALORIMETER AND PLACE BACK ON CART!